Foundations of Deep Learning

Project topics

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Project topics

- The project must be centered around one of the datasets listed in the following slides.
  - URLs marked with “(MATLAB)” are embedded in the MATLAB application. They can be downloaded and exported from one of the UNIMIB laboratories.
- Each dataset is associated with a task (classification, regression, etc.)
  - Tasks that are not covered by the course are marked in orange. You may keep these tasks at your own risk, or repurpose the dataset for a different task.
- Some datasets are particularly large, and are marked in magenta.
  - In this case you may subsample it for easier use. Please follow the “Guide to the exam” document for additional instructions on dataset subsampling.
- The students can also select a dataset not present in the list.
  - However, this must be submitted for approval to the teachers, specifying all the relevant characteristics (e.g. cardinality, dimensionality, classes, task, etc.)
- Note: anti-plagiarism software will be used to scan the submitted solutions.

Project topics 1 - image data

Dataset name	Input type	Instances	Input size	DB size	Classes	Task	URL
Omniglot	image+XYstroke	1623		5MB		similarity	https://github.com/brendenlake/omniglot
Flowers	image	3670		218MB	5	classification	http://download.tensorflow.org/example_images/flower_photos.tgz
ExampleFood	image	978		77MB	9	classification	(MATLAB)
CIFAR-10	image	60000	32x32x3	175MB	10	classification	https://www.cs.toronto.edu/~kriz/cifar-10-matlab.tar.gz
CamVid	image		720x960x3	573MB	32	segmentation	http://web4.cs.ucl.ac.uk/staff/g.brostow/MotionSegRecData/
Vehicle	image	295	720x960x3			detection	(MATLAB)
RIT-18	image			3GB	18	segmentation	https://home.cis.rit.edu/~cnspci/other/data/rit18_data.mat
BraTS	image	750	240x240x155x4	7GB		segmentation	http://medicaldecathlon.com/
Camelyon16	image	400		451GB		classification	https://camelyon17.grand-challenge.org/Data/
LowDose	image	299		1TB		img2img	https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=52758026
COCO 2014	image	82783				captioning	https://cocodataset.org/#download
IAPR TC-12	image	20000		2GB		img2img	http://www-i6.informatik.rwth-aachen.de/imageclef/resources/iaprtc12.tgz
Zurich	image	48043	448x448x3	22GB		img2img	https://data.vision.ee.ethz.ch/ihnatova/public/zr2d/Zurich-RAW-to-DSLR-Dataset.zip
SID	image			25GB		img2img	https://storage.googleapis.com/isl-datasets/SID/Sony.zip
LIVE in the Wild	image	1169				regression	https://live.ece.utexas.edu/research/ChallengeDB/index.html
MIR WM-811K	image	811457		3.5GB	10	classification	http://mirlab.org/dataSet/public/MIR-WM811K.zip
PCB Defect	image	1386		1.87GB	6	detection	https://github.com/Ironbrotherstyle/PCB-DATASET/archive/refs/heads/master.zip
BUSI	image	830		197MB	3	segmentation	(MATLAB)
Sunnybrook	image	45		105MB		segmentation	(MATLAB)
HMDB	video	7000 clips		2GB	51	classification	https://serre-lab.clps.brown.edu/resource/hmdb-a-large-human-motion-database/

Additional information and references for the listed datasets are available here:
https://www.mathworks.com/help/deeplearning/ug/data-sets-for-deep-learning.html
Orange cells constitute a warning for very large files, or tasks not covered within the course.

Project topics 2 - time series

Dataset name	Input type	N instances	DB Size	Features	Classes	Task	URL
Japanese vowels	audio			12	9	classification	(MATLAB)
Chickenpox	records					forecast	(MATLAB)
Human Activity	wearable	7 time series		3		classification	(MATLAB)
Turbofan	other signal			26		forecast	(MATLAB)
PhysioNet 2017	EEG		95MB			classification	(MATLAB)
Tennessee Eastman Process	other signal		2GB			classification	(MATLAB)
PhysioNet ECG	ECG	105 patients	72MB	2		classification	https://github.com/mathworks/physionet_ECG_segmentation
Synthetic backscattering	other signal	Inf			3	classification	(MATLAB)
Generated waveforms	other signal	3000		1	3	classification	(MATLAB)
Speech Commands	audio	65000	2GB	1	12	classification	https://storage.googleapis.com/download.tensorflow.org/data/speech_commands_v0.01.tar.gz
Mozilla Common	audio		28GB			classification	https://commonvoice.mozilla.org/
Free Spoken Digit Dataset	audio	2000			10	classification	https://github.com/Jakobovski/free-spoken-digit-dataset
Berlin Database of Emotional Speech	audio	535	40MB		7 or 10	classification	http://emodb.bilderbar.info/index-1280.html
TUT Acoustic Scenes 201	audio				15	classification	https://zenodo.org/record/400515

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Project topics 3 - text

Dataset name	Input type	N inputs	Task	URL
Factory reports	text	500 reports	classification	(MATLAB)
Shakespeare's Sonnets	text	154 sonnets	generation	(MATLAB)
ArXiv	text		classification	https://arxiv.org/help/api
Project Gutenberg	text		generation	https://www.gutenberg.org/
Weekend updates	text		classification	(MATLAB)
Finance Reports	text		classification	https://www.sec.gov/os/accessing-edgar-data

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